Sasinathan Rangasamy

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Professional Summary

Embedded Software Engineer with experience in Embedded C, firmware development, STM32, PIC16F877A, ARM Cortex-M, peripheral driver development, FreeRTOS, UART, SPI, I2C, CAN, JTAG debugging, and embedded software testing. Familiar with Git, Linux, Vector CANoe, CANalyzer, and the software development lifecycle. Seeking an opportunity in automotive embedded software and firmware development.

Technical Skills

Programming	Embedded C, C, Python
Microcontrollers	STM32, PIC16F877A, ARM Cortex-M
Protocols	UART, SPI, I2C, CAN
Embedded Systems	Firmware Development, Device Drivers, GPIO, Timers, Interrupts
RTOS	FreeRTOS
Debugging	JTAG, Firmware Debugging
Automotive Tools	Vector CANoe, Vector CANalyzer
Development Tools	MPLAB X IDE, Proteus, Arduino IDE, Git, GitHub, Jira
Operating Systems	Linux, Windows

Professional Experience

Junior Embedded Developer — **Aug 2025 – Present**
Aumovio — **Bangalore, Karnataka**

- Developed and maintained embedded firmware using **Embedded C** for STM32 and ARM Cortex-M based microcontrollers.
- Implemented and tested UART, SPI, I2C and CAN communication interfaces.
- Assisted in developing peripheral drivers for GPIO, Timers, UART and SPI modules.
- Worked with FreeRTOS for task scheduling, queues, semaphores and inter-task communication.
- Debugged firmware using JTAG and supported CAN communication testing using Vector CANoe and CANalyzer.
- Used Git, Jira and Linux throughout firmware development, testing and issue tracking.

Data Analyst (ADAS) — **Aug 2022 – Jul 2025**
Continental Autonomous Mobility — **Bangalore, Karnataka**

- Contributed to ADAS and autonomous driving projects by validating LiDAR and camera datasets.
- Developed Python scripts to automate JSON processing, data validation and report generation.
- Collaborated with engineering teams to improve dataset quality and support machine learning workflows.

Projects

Smart Irrigation System using PIC16F877A — **Academic Project**

Designed and developed a smart irrigation system using the **PIC16F877A** microcontroller and **Embedded C** to automate water pump control based on soil moisture levels. Interfaced soil moisture sensors, relay modules and a 16×2 LCD using GPIO, implemented UART communication for monitoring and debugging, and validated the firmware using **MPLAB X IDE**, **XC8 Compiler** and **Proteus**.

Education

B.E. Electronics and Communication Engineering — **2022**
Sri Sai Ranganathan Engineering College — **CGPA: 8.26/10**

Certification

Advanced Embedded Systems Certification — **2026**
Bytes In Bits Technologies — **Bangalore**